

Revision Notes for Class 8 Maths Chapter 4 Quadrilaterals

A quadrilateral is any closed figure with four sides and four angles. The name comes from the Latin words “quadri” meaning four and “latus” meaning sides. Quadrilaterals are among the basic shapes in geometry and include rectangles, squares, parallelograms, rhombuses, kites, and trapeziums. Each type has its own unique properties and uses in everyday life, such as the shapes of paper, screens, and tables.

Rectangles and Their Properties

A rectangle is defined as a quadrilateral where all four angles are right angles (90°), and the opposite sides are equal in length. Rectangles have important geometric properties.

For a rectangle:

- All angles are 90°
- Opposite sides are equal and parallel
- The diagonals are equal in length and bisect each other

If you join the corners (vertices) of a rectangle with straight lines (diagonals), each diagonal cuts the other in half, and both diagonals are always the same length.

Square: A Special Rectangle

A square is a rectangle with all sides equal. Each angle in a square is also a right angle (90°). So, every square is a rectangle, but not every rectangle is a square. In addition to rectangle properties, a square also has:

- All sides are equal
- Diagonals are equal, bisect each other at 90° , and also bisect the angles at each corner (splitting each right angle into two 45° angles)

When constructing a square using its diagonals, make sure that the diagonals are equal and they intersect at a right angle (90°).

Diagonals and Angle Properties

The diagonals of rectangles and squares always have special features. For rectangles, the diagonals are the same length and cross at their midpoints. For squares, not only

are the diagonals equal and intersect at their midpoints, but they also meet at right angles. This makes squares easy to recognize in shapes and grids.

Sum of Angles in a Quadrilateral

Regardless of the kind of quadrilateral, the sum of all its four angles is always 360° . This means if you know three angles, you can always find the fourth by subtracting their sum from 360° . This rule helps in solving many geometry questions and constructions.

Parallelograms: More About Parallel Sides

A parallelogram is another important quadrilateral. Its opposite sides are both equal and parallel, but the angles do not have to be right angles. A rectangle is a special case of a parallelogram where all the angles are 90° . Parallelograms have the following properties:

- Opposite sides are equal and parallel
- Opposite angles are equal
- Adjacent angles add up to 180°
- The diagonals bisect each other (but are not necessarily equal)

Many objects, such as picture frames and books, represent parallelograms in everyday life.

Rhombus: All Sides Equal, Angles May Differ

A rhombus is a parallelogram with all sides of equal length, but its angles are not always 90° . The special features of a rhombus include:

- All sides are equal
- Opposite sides are parallel
- Opposite angles are equal and adjacent angles add up to 180°
- Diagonals bisect each other at right angles (90°) and also bisect the angles at the corners

A square is actually a kind of rhombus where all angles are also 90° , but not all rhombuses are squares.

Kite: Unique Pairs of Sides

A kite is a quadrilateral that has two pairs of adjacent sides equal in length. The diagonal connecting the unequal angles bisects the other diagonal at 90° . Some properties of a kite:

- Two consecutive pairs of equal sides
- One diagonal bisects the other at a right angle
- A kite is not always symmetrical like a square or rectangle

The shape of actual flying kites often matches this geometric form.

Trapezium: Parallel Sides, But Not Always Equal

A trapezium (or trapezoid) is a quadrilateral with at least one pair of opposite sides parallel, called bases. The non-parallel sides are called legs. If both legs are equal, it is called an isosceles trapezium. Properties include:

- At least one pair of parallel sides
- Angles on the same side of the leg add up to 180°
- Isosceles trapeziums have equal non-parallel sides and equal base angles

Many bridges, tables, and objects have trapezium shapes for strength and design.

Relations among Quadrilaterals

A square fits into several categories: it is a rectangle (all angles 90°), a rhombus (all sides equal), and also a parallelogram (opposite sides parallel and equal). Meanwhile, rectangles and rhombuses are always parallelograms, but not all parallelograms are rectangles or rhombuses. Kites and trapeziums do not fit into other categories unless they have more specific properties.

Summary Table of Quadrilateral Properties

- **Rectangle:** All angles 90° , opposite sides equal and parallel, diagonals equal and bisect each other
- **Square:** All angles 90° , all sides equal, diagonals equal, bisect at 90° , bisect each angle

- **Parallelogram:** Opposite sides parallel and equal, opposite angles equal, adjacent angles add to 180° , diagonals bisect each other
- **Rhombus:** All sides equal, opposite sides parallel, diagonals bisect at 90° , diagonals bisect angles, adjacent angles add to 180°
- **Kite:** Two pairs of equal adjacent sides, one diagonal bisects the other at a right angle
- **Trapezium:** At least one pair of opposite sides is parallel
- **Angle Sum:** For all quadrilaterals, the sum of angles is 360°

Activities and Applications

You can explore quadrilaterals by folding sheets, joining triangles, and using geoboards. For example, joining two equilateral triangles can form a rhombus or kite based on the arrangement. If four equal sides and one right angle are present, the quadrilateral must be a square. Understanding the definitions and differences between these figures helps in classifying shapes in geometry problems.

In summary, quadrilaterals have various forms and each type has unique properties related to angles, sides, and diagonals. Recognizing these properties makes geometry simpler and more logical, especially when solving problems or working on constructions.